

Solution Requirements

Date	15 March 2026
Team ID	XXXXXX
Project Name	XXXXXX
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>[Describe how users will securely log in or verify their identity]</i>	
2	Authorization levels	<i>[Define the user roles and what access/permissions each role has]</i>	
3	External interfaces	<i>[Detail any third-party APIs, external databases, or</i>	

		<i>services the system must connect to]</i>	
4	Transactions processing	<i>[Explain how data exchanges, payments, or core system transactions will be handled]</i>	
5	Reporting	<i>[Describe the types of reports, dashboards, or data exports the system must generate]</i>	
6	Business rules, etc.	<i>[List specific logic, constraints, or workflows unique to the business process]</i>	
7	Compliance to laws or regulations	<i>[Specify any legal, industry, or data privacy standards the system must adhere to (e.g., GDPR, HIPAA)]</i>	
8	<i>Other</i>	<i>[Add any additional functional requirements here]</i>	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>[Describe response times, processing speed, etc.]</i>	<i>(e.g., Page loads in < 2 seconds)</i>
2	Scalability	<i>[Describe how the system handles growth in users or data]</i>	<i>(e.g., Supports up to 10,000 concurrent users)</i>
3	Security & Data Privacy	<i>[Detail data protection, encryption, and vulnerability management]</i>	<i>(e.g., End-to-end AES-256 encryption)</i>
4	Reliability & Availability	<i>[Specify uptime requirements and fault tolerance]</i>	<i>(e.g., 99.9% uptime per month)</i>
5	Usability & Accessibility	<i>[Describe user experience standards and accessibility compliance]</i>	<i>(e.g., Complies with WCAG 2.1 AA standards)</i>
6	Other	<i>[Add any additional non-functional requirements (e.g., Maintainability, Portability)]</i>	

SOLUTION REQUIREMENTS

Functional Requirements:

- Model must take 7 parameters: Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall.
- System must return a single optimal crop string via a web interface.

Non-Functional Requirements:

- Prediction API must respond in under 2 seconds.
- Machine Learning model accuracy must exceed 90%.